

2025 Invitational Tournament - Div C

Machines Exam



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School Name/Team: _____

Student Names: _____

Instructions:

1. Write all answers on the Answer Sheet. Do not write on this Test Packet.
2. Points are based on accurately answering each part of the question.
3. Ties marked with a * will be broken in order
4. This test has a total of 24 points.

1. A 60 N load is placed 0.3 m from the pivot of a lever. To lift it, an effort of 15 N is applied at a distance of 0.9 m on the opposite side. Which statement is correct? [1]
 - a. The lever is balanced
 - b. The lever rotates clockwise
 - c. The lever rotates counterclockwise
 - d. The lever is in equilibrium

Use the following word bank for questions 2-8. You can repeat words. [One point for each]

A. First Class Lever	B. Second Class Lever	C. Third Class Lever	D. None of the Above
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2. A crowbar: [1] _____
3. A fixed pulley on the ceiling: [1] _____
4. A pair of tweezers: [1] _____
5. A wheelbarrow: [1] _____
6. A nutcracker: [1] _____
7. A human arm lifting a weight: [1] _____
8. A dishing rod: [1] _____
9. The tangent of the angle of repose equals: [1]
 - a. Load / Effort
 - b. Height / Length
 - c. Coefficient of static friction
 - d. Efficiency
10. A higher coefficient of friction will... [1]
 - a. Decrease the angle of repose
 - b. Increase the angle of repose
 - c. Leave it unchanged
 - d. Make the surface frictionless
11. The mechanical advantage of a screw is proportional to: [1]
 - a. The diameter of the screw
 - b. The pitch of the thread
 - c. The number of turns per unit length
 - d. Both A and B
 - e. Both A and C
12. A second-class lever always has: [1]
 - a. The fulcrum between the load and the effort
 - b. The load between the fulcrum and the effort
 - c. The effort between the load and the fulcrum
 - d. Equal effort and load

13. In a pulley system with ideal mechanical advantage (IMA) = 4, an effort of 200 N lifts a 700 N load. What is the actual mechanical advantage (AMA)? [1]
- 2.5
 - 3.0
 - 3.5
 - 4.0
14. In a third-class lever, the effort is applied 0.5 m from the fulcrum, while the load is 2.0 m away. If the effort force is 120 N, what is the load force ideally? [1]
- 25 N
 - 30 N
 - 40 N
 - 50 N
15. A wheel radius is 0.3 m, and the axle radius is 0.05 m. A 50 N effort is applied at the wheel. What force is applied at the axle ideally? If the efficiency is 80%, what is the actual force? [1*]
- 300 N; 240 N
 - 300 N; 250 N
 - 200 N; 160 N
 - 250 N; 200 N
16. Which of the following best defines the Ideal Mechanical Advantage (IMA) of a simple machine? [1]
- The ratio of input force to output force
 - The ratio of load distance to effort distance
 - The ratio of actual mechanical advantage to efficiency
 - The product of efficiency and load
17. A first-class lever is in equilibrium. If the input force acts 0.3 m from the fulcrum and the output force acts 0.1 m on the opposite side, which statement is true? [1]
- The input force equals the output force.
 - The input force is one-third of the output force.
 - The input force is three times the output force.
 - The lever provides no mechanical advantage.
18. A 2 kg box slides down a 2 m long frictionless incline with vertical height 0.5 m. Find the speed at the bottom of the incline. [1*]
- 1.5 m/s
 - 2.2 m/s
 - 3.1 m/s
 - 4.0 m/sd

19. In a frictionless inclined plane system at rest, which type of energy is primarily stored in the object placed on the ramp? [1]
- Kinetic Energy
 - Elastic Potential Energy
 - Chemical Energy
 - Gravitational Potential Energy
20. A block is lifted by a pulley system with 3 supporting ropes. The system has a frictional efficiency of 80%. If the load is 360 N, what effort is needed? [1]
- 120 N
 - 150 N
 - 180 N
 - 200 N
21. Sand forms a pile with a slope angle of 30° . What is the coefficient of static friction between sand grains? [1]
- 0.50
 - 0.58
 - 0.33
 - 0.69
22. A student claims that all second-class levers are self-locking. Which statement explains why this is incorrect? [1]
- Second-class levers always multiply speed
 - Only screws and wedges can be self-locking
 - Self-locking is impossible in any lever
 - Self-locking depends on friction, not lever class
23. A wedge is hammered into a wooden block at an angle of 12° ; the friction coefficient is 0.1. What happens if the block is vertical and you remove your hammer? [1*]
- The wedge stays in place
 - The wedge slides down
 - Mechanical advantage doubles
 - Mechanical advantage decreases by half
24. The effort force for an incline without friction equals (where m is the mass of the object and g is the acceleration due to gravity): [1]
- $mg \sin \theta$
 - $mg \cos \theta$
 - $mg / \sin \theta$
 - $mg \tan \theta$